

Jay V. Kalinani

Postdoctoral Researcher | University of Illinois Urbana-Champaign

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Research profile. Computational relativistic astrophysicist developing GPU-accelerated GRMHD and neutrino-radiation codes for compact-object mergers, with an emphasis on connecting exascale simulations to multimessenger observables.

Academic Appointments

Aug 2025–present	Postdoctoral Researcher , University of Illinois Urbana-Champaign, USA Advisor: Prof. Helvi Witek
Jan 2023–Jul 2025	Postdoctoral Researcher , Center for Computational Relativity and Gravitation, Rochester Institute of Technology, USA Advisor: Prof. Manuela Campanelli
Jan–Dec 2022	Postdoctoral Researcher , INAF–Astronomical Observatory of Padova, Italy Advisor: Dr. Riccardo Ciolfi

Education

2018–2021	Ph.D. in Physics , University of Padova, Italy, <i>summa cum laude</i> Thesis: Binary Neutron Star Mergers in the Era of Multi-Messenger Astrophysics Advisors: Dr. Riccardo Ciolfi and Prof. Jean-Pierre Zendri
2016–2018	M.S. in Astrophysics , AstroMundus Programme , 110/110 cum laude Thesis: GRMHD Simulations of Binary Neutron Star Mergers
2014–2016	M.S. in Applied Solid State Physics , M. S. Ramaiah University of Applied Sciences, India, CGPA 8.17/10 Thesis: <i>Acoustic Scattering by Different Geometrical Structures</i>
2010–2014	B.Tech. in Nuclear Science & Engineering , SRM University, India, CGPA 9.01/10 Thesis: <i>Design Aspects of a Sodium Heated Steam Generator</i>

Selected Research Contributions

- **Exascale relativistic astrophysics.** Led development of AsterX, an open-source GPU-accelerated GRMHD code for dynamical spacetimes, validating it across standard benchmarks and scaling it to 4,096 nodes on Frontier. Leads complementary GPU software for neutrino transport, horizon finding, and particle tracing. (*Kalinani et al., 2025*)
- **Relativistic outflows from neutron-star mergers.** Led simulations following two sequential polar outflows within one merger: first a dense massive-neutron-star-driven outflow and later a black-hole-disk jet compatible with a short gamma-ray burst. The simulations substantially reduced artificial-atmosphere effects and resolved the interaction between the outflows. Earlier work connected magnetically driven remnant winds to the blue kilonova of GW170817. (*Ciolfi & Kalinani, 2020; Kalinani et al., 2026*)
- **Microphysics and nucleosynthesis.** Contributed to merger simulations incorporating temperature-dependent nuclear equations of state, neutrino transport, and neutron-star spin, quantifying their effects on magnetic-field amplification, polar outflows, ejecta composition, and kilonova signatures. (*Nouri et al., 2026; Wen et al., 2026*)
- **Supermassive black-hole binaries.** Led simulations of a precessing, unequal-mass binary that connected disk crossings, jet precession, and intermittent jet weakening to quasi-periodic electromagnetic signatures; also contributed to inspiral-through-merger accretion simulations predicting a transient luminosity enhancement. (*Ennoggi et al., 2025; Kalinani et al., 2026*)
- **Jet propagation across scales.** Contributed to cross-code handoffs from GRMHD merger remnants to large-scale relativistic jet simulations, linking central-engine and ejecta conditions to jet breakout and observable signatures. (*Pavan et al., 2021, 2023, 2025*)

Research Software and Infrastructure

Lead developer of GPU-accelerated research software for numerical relativity, GRMHD, neutrino transport, and particle-based analysis, primarily within the Einstein Toolkit ecosystem.

Lead developer and maintainer

2022–present	AsterX – GPU-accelerated GRMHD for dynamical spacetimes; demonstrated scaling to 4,096 Frontier nodes.
2024–present	nuX – GPU-accelerated moment-based neutrino radiation transport.
2026–present	BHClusterX – Initial-data tools for black-hole cluster simulations in the Einstein Toolkit.
2026–present	FlowTracerX – GPU-accelerated passive fluid-particle tracing for AsterX.

- 2026–present **ParticleAHFinderX** – GPU apparent-horizon finding with AMReX particles.
 2025–present **AsterAI** – AI-assisted code-review tools for numerical relativity.

Developer and contributor

- 2020–present **Spritz** – GRMHD with neutrino leakage in fully dynamical spacetimes; developer.
 2022–present **SpacetimeX** – Modules for spacetime initialization and evolution; developer.
 2022–present **CarpetX** – AMReX-based Cactus driver for the Einstein Toolkit; contributor.
 2021 **RePrimAnd** – Primitive-variable recovery and equation-of-state framework for GRMHD; contributor.

Research Funding and Computing Allocations

Research funding

- 2023–2025 **Co-Investigator**, JetOut: “Breakout, propagation, and observational signatures of short gamma-ray burst jets from binary neutron star mergers.” Italy–USA scientific-cooperation project funded by the Italian Ministry of Foreign Affairs and International Cooperation; PI: Riccardo Ciolfi. **Total award: €131,000.**
 Apr–Jul 2016 **CEFIPRA Research Grant**, Aquitaine–Karnataka pre-Ph.D. program. **Award: €4,500.**

High-performance computing

PI or Co-I on 27 allocations totaling more than 50.4 million CPU-hours and 1 million GPU-hours. **PI on 10 allocations** totaling approximately 15.5 million CPU-hours and 496,000 GPU-hours; PI awards are listed below.

- Aug 2026–Feb 2027 **ALCF Director’s Discretionary, Polaris** – 5,000 GPU-hours. *Scaling and Validation of Black-Hole Cluster Simulations.*
 Mar 2026–Mar 2027 **ACCESS Discover, DeltaAI** – 11,000 GPU-hours. *GPU validation and benchmarking of AsterX/CarpetX with Co-callIDX initial data.*
 Oct 2025–Oct 2026 **OLCF Director’s Discretionary, Frontier** – 160,000 GPU-hours. *Benchmarking AsterX and CanudaX with magnetized neutron-star mergers.*
 Feb 2025–Feb 2026 **TACC Pathways, Frontera** – 12.6 million CPU-hours. *Relativistic jet formation and bulk viscosity in magnetized neutron-star mergers.*
 May 2023–May 2025 **OLCF Director’s Discretionary, Frontier** – 320,000 GPU-hours. *Benchmarking the GPU-accelerated GRMHD code AsterX.*
 Dec 2022–Jun 2023 **INAF–CINECA Class B, Galileo 100** – approximately 375,000 CPU-hours. *Binary neutron-star mergers with Spritz and the APR4 equation of state.*
 Jul–Dec 2022 **EuroHPC Benchmark Access, Discoverer** – approximately 1 million CPU-hours. *Binary neutron-star merger simulations with Spritz.*
 Mar–Dec 2022 **CINECA Class C, Galileo 100** – approximately 100,000 CPU-hours. *Magnetized binary neutron-star merger simulations with Spritz.*
 Aug 2021–May 2022 **CINECA Class C, Galileo 100** – approximately 192,000 CPU-hours. *Fishbone–Moncrief disk simulations with Spritz.*
 Apr 2020–Apr 2021 **CINECA Class B, Marconi 100 and Galileo** – approximately 1.25 million CPU-hours. *Binary neutron-star merger simulations with Spritz.*

Publications

18 papers: 15 published, 1 accepted, and 2 submitted; more than 550 citations; h-index 12. Citation metrics from NASA ADS, September 2026. The candidate’s name is shown in bold.

Submitted manuscripts

- 2026 **Jet variability in small mass ratio supermassive black hole binaries.** [first author]
J. V. Kalinani, M. Campanelli, M. Chabanov, M. C. de Simone, C. O. Lousto, H. Witek, and Y. Zlochower.
Submitted to ApJL. arXiv:2609.24837.
 2026 **Binary Neutron Star Merger Simulations with Microphysical Equation of State using Spritz.**
 F. H. Nouri, **J. V. Kalinani**, B. Giacomazzo, R. Ciolfi, and A. Perego.
Submitted to CQG. arXiv:2608.23310.

Accepted for publication

- 2026 **Magnetic eruption and nucleosynthesis in GRMHD simulations of spinning neutron star mergers.**
 A. Wen, **J. V. Kalinani**, M. Chabanov, M. Campanelli, R. Ciolfi, and Y. Zlochower.
Accepted for publication in PRD. arXiv:2605.30548.

Published peer-reviewed articles

- 2026 **Jet-environment interaction after delayed collapse in binary neutron star mergers.** [first author]
J. V. Kalinani, R. Ciolfi, M. Campanelli, B. Giacomazzo, A. Pavan, A. Wen, and Y. Zlochower.
ApJL 1000, L35.
 2026 **Accelerating numerical relativity simulations with new multistep fourth-order Runge–Kutta methods.**
 L. T. Sanches, S. R. Brandt, **J. V. Kalinani**, L. Ji, and E. Schnetter.
CQG 43, 065010.

- 2025 **Relativistic gas accretion onto supermassive black hole binaries from inspiral through merger.**
L. Ennoggi, M. Campanelli, Y. Zlochower, S. C. Noble, J. Krolik, F. Cattorini, J. V. Kalinani, V. Mewes, M. Chabanov, L. Ji, and M. C. de Simone.
[PRD 112, 063009.](#)
- 2025 **Role of injection parameters in jet propagation through realistic binary neutron star merger environments.**
A. Pavan, R. Ciolfi, E. Dreas, and J. V. Kalinani.
[MNRAS 540, 1345–1362.](#)
- 2025 **AsterX: a new open-source GPU-accelerated GRMHD code for dynamical spacetimes.** [\[first author\]](#)
J. V. Kalinani, L. Ji, L. Ennoggi, F. G. Lopez Armengol, L. T. Sanches, B. J. Tsao, S. R. Brandt, M. Campanelli, R. Ciolfi, B. Giacomazzo, R. Haas, E. Schnetter, and Y. Zlochower.
[CQG 42, 025016.](#)
- 2023 **Jet-environment interplay in magnetized binary neutron star mergers.**
A. Pavan, R. Ciolfi, J. V. Kalinani, and A. Mignone.
[MNRAS 524, 260–275.](#)
- 2022 **Implementing a new recovery scheme for primitive variables in the GRMHD code Spritz.** [\[first author\]](#)
J. V. Kalinani, R. Ciolfi, W. Kastaun, B. Giacomazzo, F. Cipolletta, and L. Ennoggi.
[PRD 105, 103031.](#)
- 2022 **Turbulent magnetic field amplification in binary neutron star mergers.**
C. Palenzuela, R. Aguilera-Miret, F. Carrasco, R. Ciolfi, J. V. Kalinani, W. Kastaun, B. Miñano, and D. Viganò.
[PRD 106, 023013.](#)
- 2021 **Short gamma-ray burst jet propagation in binary neutron star merger environments.**
A. Pavan, R. Ciolfi, J. V. Kalinani, and A. Mignone.
[MNRAS 506, 3483–3498.](#)
- 2021 **Spritz: General Relativistic Magnetohydrodynamics with Neutrinos.**
F. Cipolletta, J. V. Kalinani, E. Giangrandi, B. Giacomazzo, R. Ciolfi, L. Sala, and B. Giudici.
[CQG 38, 085021.](#)
- 2021 **Robust Recovery of Primitive Variables in Relativistic Ideal Magnetohydrodynamics.**
W. Kastaun, J. V. Kalinani, and R. Ciolfi.
[PRD 103, 023018.](#)
- 2020 **Magnetically driven baryon winds from binary neutron star merger remnants and the blue kilonova of August 2017.**
R. Ciolfi and J. V. Kalinani.
[ApJL 900, L35.](#)
- 2020 **Spritz: a new fully general-relativistic magnetohydrodynamic code.**
F. Cipolletta, J. V. Kalinani, B. Giacomazzo, and R. Ciolfi.
[CQG 37, 135010.](#)
- 2019 **First 100 ms of a long-lived magnetized neutron star formed in a binary neutron star merger.**
R. Ciolfi, W. Kastaun, J. V. Kalinani, and B. Giacomazzo.
[PRD 100, 023005.](#)
- 2014 **Thermal Analysis of a Mobile Hot Cell Cask.** [\[first author\]](#)
J. V. Kalinani and A. R. Kumar.
[Int. J. Innov. Res. Sci. Eng. Technol. 3\(11\), 17575–17584.](#)

Teaching Experience

UIUC	Guest lecturer, undergraduate Classical Mechanics – lectures on drag forces and harmonic oscillators. Course instructors: Profs. Hector O. Silva and Helvi Witek.
RIT	Guest lecturer, graduate astrophysics – lectures on general relativistic magnetohydrodynamics and numerical methods for conservation laws in <i>Astrophysical Flows</i> and <i>Fundamentals of Theoretical Astrophysics I</i> . Course instructor: Prof. Manuela Campanelli.
Einstein Toolkit	Workshop instructor – designed and delivered hands-on tutorials on CarpetX and AsterX at community workshops and schools.

Research Mentoring

Graduate students	Co-mentor: Allen Wen (RIT, Dec 2023–present); Amara Nusbickel (RIT, Mar 2025–Mar 2026); Maxwell Rizzo (UIUC, Aug 2025–present); Noora Ghadiri (UIUC, Mar 2026–present); Freddy Par-doe (UIUC, Mar 2026–present).
Hackathons	Mentor: Bing-Jyun Tsao (2023) and Maxwell Rizzo (2025) during Einstein Toolkit hackathons.

Honors and Awards

2018–2021	CARIPARO Ph.D. Scholarship , Cariparo Foundation, Italy (€99,800).
Apr 2019	Best Ph.D. Student Talk , PHAROS Conference, Playa de Aro, Spain.
2016–2018	AstroMundus Consortium Scholarship (€43,400).
Oct 2016	Best Master’s Thesis Award , M.S. Ramaiah University, India.
2010–2012	Department Scholarship for Rank 1 , SRM University, India.
2007–2008	AFS Kennedy–Lugar Youth Exchange and Study Scholarship , one-year high-school exchange in San Diego, USA.

Selected Professional Development

Jul–Aug 2026	Argonne Training Program on Extreme-Scale Computing – selected intensive program in exascale application design, implementation, and execution.
Sep 2025– Apr 2026	Scholarship and Teaching for Engineering Postdocs Program – faculty-preparation program in teaching, mentoring, and research.

Academic Service and Research Leadership

Organization	Local and scientific organizer, North American Einstein Toolkit Workshop and School (UIUC, 2026); local organizer, CSSI Einstein Toolkit Hackathon (2025); organizer, CCRG Friday lunch seminars.
Leadership	Team lead, AsterX/CarpetX group, Frontier Hackathon (2023).
Peer review	Reviewer for the U.S. National Science Foundation, Physical Review journals, <i>Classical and Quantum Gravity</i> , and <i>Astronomy & Astrophysics</i> .
Memberships	American Physical Society (DGRAV and DCOMP); LISA Consortium; Theory and Computational Astrophysics Network; Indian Association for General Relativity and Gravitation.

Presentations

46 presentations: 23 invited talks, seminars, and tutorials; 23 contributed talks and posters.

Selected invited presentations

Jul 2026	Relativistic outflows from compact-object binaries. New Frontiers in Strong Gravity, Benasque, Spain.
Mar 2026	General relativistic magnetohydrodynamics on GPUs. APS Global Physics Summit, Denver, USA.
Feb 2026	Towards end-to-end modelling of jets from BNS mergers. Nuclear Physics from Multi-Messenger Mergers Seminar, online.
Apr 2025	End-to-end modelling of jets from BNS mergers. The 7th Neighbourhood Workshop, Penn State University, USA. [plenary]
Oct 2024	End-to-end modelling of jets from BNS mergers. 130th Topical Symposium, New York State Section of the APS, SUNY Polytechnic Institute, USA.
Jul 2024	AsterX: a new open-source GPU-accelerated GRMHD code for dynamical spacetimes. Solving the Boltzmann Equation for Neutrino Transport, ICERM, USA.
Aug 2023	Binary Neutron Star Merger Simulations: Long-Lived Remnants, Short Gamma-Ray Bursts & Kilonovae. INT 23-2 Workshop, University of Washington, USA.
May 2023	Binary Neutron Star Merger Simulations: Long-Lived Remnants, Short Gamma-Ray Bursts & Kilonovae. Numerical Relativity Seminar, online.

Additional invited talks, seminars, and tutorials

Dec 2025	Tutorial on the AsterX code. European Einstein Toolkit Meeting, online. [tutorial]
Oct 2025	End-to-end modelling of jets from BNS mergers. Astrophysics, Relativity, and Cosmology Seminar, UIUC, USA.
Aug 2025	nuX-M1: a new GPU-accelerated moment-based neutrino radiation transport code. Einstein Toolkit CSSI Workshop, UIUC, USA.
Jun 2025	The AsterX and nuX code. North American Einstein Toolkit Workshop, University of Texas at Austin, USA.
Oct 2024	End-to-end modelling of jets from BNS mergers. NEOSGrav Conference, Goa, India.
Jul 2024	AsterX: a new open-source GPU-accelerated GRMHD code for dynamical spacetimes. North American Einstein Toolkit Workshop, Louisiana State University, USA.
Jul 2023	Introduction to GRMHD and the AsterX code. North American Einstein Toolkit School, RIT, USA.
Jul 2023	The CarpetX/AsterX code. North American Einstein Toolkit School, RIT, USA. [tutorial]
Oct 2022	Magnetized BNS merger simulations and prospects of launching SGRB jets. EMMI–IReNA GSI Workshop, online.
Jun 2022	Binary Neutron Star Merger Simulations with WhiskyMHD and Spritz. TCAN Meeting, online.
Jun 2022	The CarpetX Framework. North American Einstein Toolkit Workshop, online. [tutorial]
Jun 2022	The Spritz Code: GRMHD with Neutrinos. North American Einstein Toolkit Workshop, online.
Jul 2021	Magnetized BNS Merger Simulations with Long Post-Merger Evolution. TCAN BNS/BH–NS Merger Workshop.
Jul 2021	Binary Neutron Star Merger Simulations: Long-Lived Remnant, Short Gamma-Ray Bursts, and Kilonovae. 16th Marcel Grossmann Meeting, online.
Jun 2021	GRMHD Simulations of Binary Neutron Star Mergers. ECT* Workshop: Neutron Stars as Multi-Messenger Laboratories for Dense Matter.

Contributed talks and posters

Jul 2026	Jet variability in small mass ratio supermassive black-hole binaries. UK & Ireland Discs Meeting, online. [poster]
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Contributed talks and posters (continued)

Jul 2026	Jet variability in small mass ratio supermassive black-hole binaries. 16th International LISA Symposium, College Park, USA.
Mar 2026	Jet-environment interaction after delayed collapse in BNS mergers. 3rd International Online Conference on Universe, online.
Dec 2024	AsterX: a new open-source GPU-accelerated GRMHD code for dynamical spacetimes. Numerical Relativity Community, online.
Apr 2024	End-to-end modelling of jets from binary neutron star mergers. APS April Meeting, Sacramento, USA.
Apr 2023	AsterX: a new open-source GPU-accelerated GRMHD code for dynamical spacetimes. APS April Meeting, Minneapolis, USA.
Jul 2022	Magnetized BNS merger simulations and prospects of launching SGRB jets. COSPAR 44th Scientific Assembly, Athens, Greece.
May 2022	Binary Neutron Star Merger Simulations with GRMHD code Spritz. PHAROS Conference, Rome, Italy.
Apr 2022	Binary Neutron Star Merger Simulations with GRMHD code Spritz. APS April Meeting, online.
Nov 2021	Magnetars born in BNS mergers, Short GRBs and Kilonovae. IAU Symposium 363, online.
Jul 2021	A new primitive variable recovery scheme in GRMHD code Spritz. North American Einstein Toolkit Summer School.
Apr 2021	The Spritz Code: Improving the Robustness of GRMHD. APS April Meeting, online.
Feb 2021	A new primitive variable recovery scheme in GRMHD code Spritz. PHAROS-CompOSE Conference, online.
Feb 2021	Binary Neutron Star Merger Simulations: Long-Lived Remnant, Short Gamma-Ray Bursts and Kilonovae. TEONGRAV Workshop, online.
Dec 2020	Role of Magnetic Fields in Binary Neutron Star Merger Simulations. IAGRG Meeting, online.
Jul 2020	Role of Magnetic Fields in Binary Neutron Star Merger Simulations. TCAN Binary Neutron Stars Workshop.
Apr 2020	GRMHD Simulations of BNS Mergers: 100 ms of Long-Lived Remnant Evolution. APS April Meeting, online.
Sep 2019	GRMHD Simulations of BNS Mergers: 100 ms of Long-Lived Remnant Evolution. European Einstein Toolkit Meeting, London, UK.
Apr 2019	GRMHD Simulations of BNS Mergers: 100 ms of Long-Lived Remnant Evolution. PHAROS Conference, Playa de Aro, Spain.
Mar 2019	GRMHD Simulations of BNS Mergers. PHAROS Ph.D. School, University of Jena, Germany. [poster]
Jan 2019	GRMHD Simulations of BNS Mergers. IAGRG Meeting, BITS Pilani Hyderabad, India. [poster]
Jun 2018	GRMHD Simulations of BNS Mergers. AstroMundus Retreat, Obergurgl, Austria. [poster]
Aug 2015	Precision Particle Tracking. Center for Axion and Precision Physics, KAIST, South Korea. [poster]

Public Engagement

2020–present	Volunteer, Skype a Scientist – interactive astronomy sessions with classrooms and families.
Apr 2023	Exhibitor, Imagine RIT: Creativity and Innovation Festival – represented the astrophysics department.
May 2020	Organizer and speaker, Let's Talk Astronomy – online public talk, <i>Galaxies, Stars and Planets</i> .
Apr 2020	Contributor, Astrobites – <i>Short Gamma Ray Burst Central Engines: A Curious Case of Time-Reversal</i> .

Additional Community Service

2023–2025	Dog-care volunteer, Rochester Animal Services, Rochester, USA.
2020–2025	Active listener, 7 Cups.
2014–2016	Education-support volunteer, Make A Difference, Bangalore, India.
2012–2014	Head of Operations and Co-Director, Project Book-Wings, The Green Nest, Chennai, India.
2012–2013	Volunteer, Child Rights and You, Bangalore and Chennai, India.

Academic References

Current advisor	Prof. Helvi Witek , University of Illinois Urbana-Champaign hwitek@illinois.edu
Former advisor	Prof. Manuela Campanelli , Center for Computational Relativity and Gravitation, Rochester Institute of Technology mxcsma@rit.edu
Ph.D./postdoc advisor	Dr. Riccardo Ciolfi , INAF–Astronomical Observatory of Padova and INFN Padova riccardo.ciolfi@inaf.it